

Waveform Synthesis Using a Multistage Op-Amp Function Generator

Lily Nguyen

December 6, 2025

1 Introduction and Theory

This project aims to design and build function generator capable of producing square, triangle, and sine waves using operational amplifiers and discrete components. Rather than relying on commercial waveform ICs, we implemented each function block separately to produce these waveforms and explore principles of signal synthesis.

Our circuit consists of three main stages: a square wave oscillator, a triangle wave integrator, and a sine wave shaper, as shows in Figure 1.

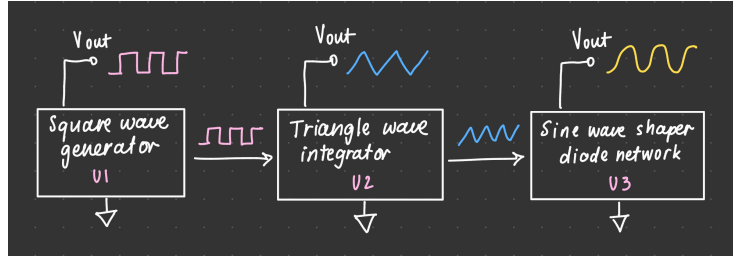


Figure 1: Block diagram of the function generator circuit. The signal flows the comparator-based square wave oscillator to an op-amp integrator that outputs a triangle wave, and finally to a nonlinear shaping circuit that produces a sine wave.

The first stage uses a comparator configuration (U1) with positive feedback via a resistor-divider network. The op-amp toggles between the positive and negative supply rails whenever the inverting input crosses the threshold voltage:

$$V_{\text{threshold}} = \pm \left(\frac{R_3}{R_1 + R_3} \right) V_{\text{out}}, \quad (1)$$

where R_1 and R_3 form the voltage divider [2]. The resistor R_2 and the capacitor in the following integrator stage define the RC time constant that controls the

output frequency. This configuration functions similarly to a Schmitt trigger, which produces a square wave by switching states as the capacitor charges and discharges [6]. The charging path through R_2 and the feedback from the integrator ensure the timing interval stays consistent and results in a stable waveform with well-defined frequency and symmetry.

The square wave generated by the comparator stage feeds into an inverting integrator composed of an op-amp (U2), resistor R_2 , and feedback capacitor C_1 . This circuit operates as an integrator by converting the constant high and low levels of the square wave into a linearly increasing or decreasing voltage [6]. The resulting output is a triangle wave where V_{out} evolves according to the relation:

$$V_{\text{out}}(t) = -\frac{1}{RC} \int V_{\text{in}}(t) dt \quad (2)$$

Increasing the square wave amplitude increases the rate of change, whereas increasing the RC constant slows it down. Assuming symmetrical saturation and ideal integration, the frequency of oscillations is $f = \frac{1}{T} = \frac{R_1}{4R_2R_3C_1}$ [2].

To convert the triangle wave into a sine-like waveform, we use a nonlinear shaping circuit composed of diodes and resistor. The diodes begin conducting at specific voltage thresholds (around 0.6-0.7 V), which alters the waveform slope in a piecewise linear manner. As the triangle wave amplitude increases, more diodes enter conduction to successively compress the waveform peaks and round them to a better approximate sine wave. The shaped signal is then passed through a summing amplifier, which allows for additional gain control and centering of the output waveform around 0 V [3].

Figure 2 shows the full circuit schematic for our function generator.

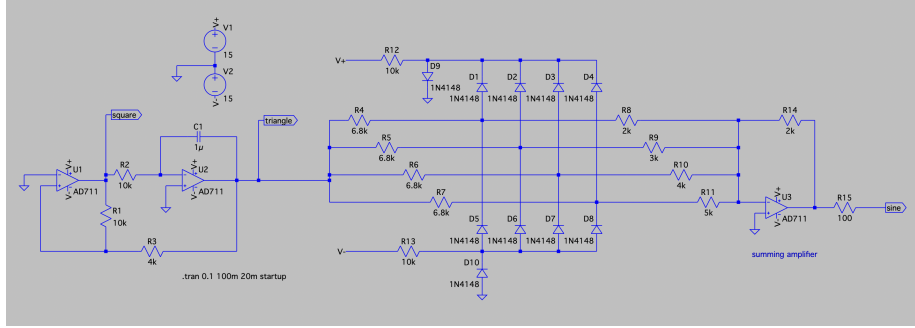


Figure 2: Full schematic of the function generator circuit. U1 forms a square wave generator using a comparator with positive feedback. U2 is configured as an inverting integrator to generate a triangle wave. U3 shapes the triangle wave into a sine wave using a diode-resistor nonlinear network and summing amplifier configuration. Component values were selected to set the operating frequency to approximately 60 Hz.

To analyze the quality of the generated sine wave, we performed Fourier analysis to decompose the waveform into its frequency components. The Discrete Fourier Transform (DFT) expresses a sampled signal $v(t_n)$ as a sum of weighted complex exponentials:

$$V_k = \sum_{n=0}^{N-1} v(t_n) e^{-2\pi i k n / N}, \quad (3)$$

where V_k is the complex amplitude of the k -th [1]. The magnitude spectrum $|V_k|$ indicates the strength of each frequency component, with the dominant peak corresponding to the fundamental frequency and additional peaks representing harmonic distortion. Any deviation from a pure sine wave introduces higher-frequency harmonics into the signal [8][5]. To quantify this distortion, we compute the Total Harmonic Distortion (THD), which measures how much of the distortion is due to harmonics in the signal:

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1^2}. \quad (4)$$

2 Methods and Results

2.1 Methods

We built the circuit from Figure 2 on a solderless breadboard powered using the supply from a PB-505A set at ± 18 V. We observed the output waveforms using a Tektronix TBS10000C 2-Channel Oscilloscope and used the **Measure** function to record the amplitude, frequency, and rise/fall times and duty cycle for the square wave. For the triangle wave, we visually checked for linearity and

symmetry.

To evaluate the frequency content of the sine wave, we performed a Fast Fourier Transform (FFT) on the waveform data using Python. We processed the data using the `fft` function from the `scipy.fftpack` package to compute the one-sided FFT magnitude spectrum. We identified the dominant frequency by locating the peak (excluding DC) and the amplitudes at the 2nd-5th harmonics to compute the Total Harmonic Distortion (THD) using Equation 4 [5]. We also performed a least-squares fit to the form:

$$V(t) = A \sin(2\pi ft + \phi) + V_0 \quad (5)$$

using the `curve_fit` function from the `scipy.optimize` package, which allowed us to extract the amplitude A , frequency f phase ϕ , and vertical offset V_0 of the signal. From these parameters, we computed the RMS voltage of the waveform.

Before constructing the circuit, we ran transient simulations of each stage using LTSpice to verify the expected waveform behavior (see Figure 3).

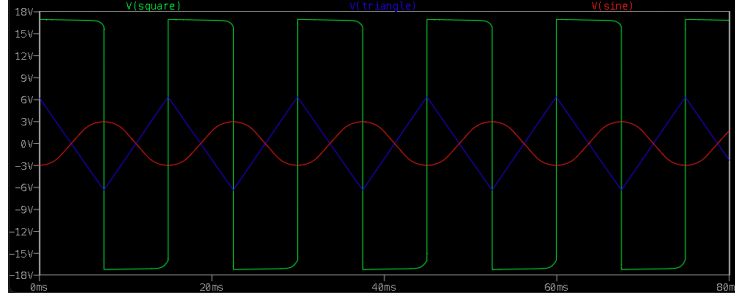


Figure 3: Transient simulation showing the output waveforms at each stage of the function generator circuit.

2.2 Results

To verify the performance of our function generator, we recorded the output waveforms of all three stages, as shown in Figure 4. Our results agree with our LTSpice simulations. The square wave exhibited sharp transitions with measured rise and fall times of approximately 25.75 μ s and 24.60 μ s, and near-symmetric duty cycles of 49.25% (positive) and 50.74% (negative), which is expected from a properly biased comparator [6]. The triangle showed linear charging and discharging behavior with consistent amplitude and frequency matching the square wave.

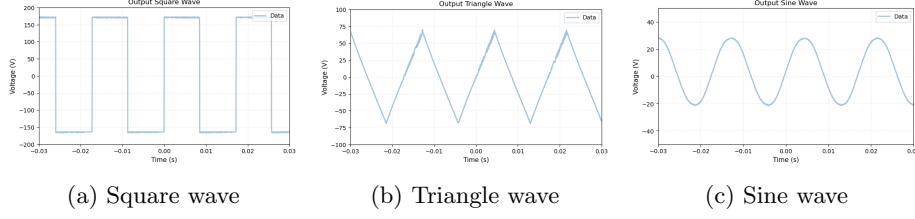


Figure 4: Oscilloscope waveforms from each stage of the function generator circuit ($f = 58.08$ Hz).

We identified the fundamental frequency from the FFT spectrum as 62.5 Hz, with secondary peaks at 125.0 Hz, 187.5 Hz, 250.0 Hz, and 312.5 Hz corresponding to the 2nd-5th harmonics. We computed the Total Harmonic Distortion to be 11.44%, which suggests our passive diode network had reasonable sine shaping performance [8]. The FFT magnitude spectrum is shown in Figure 5.

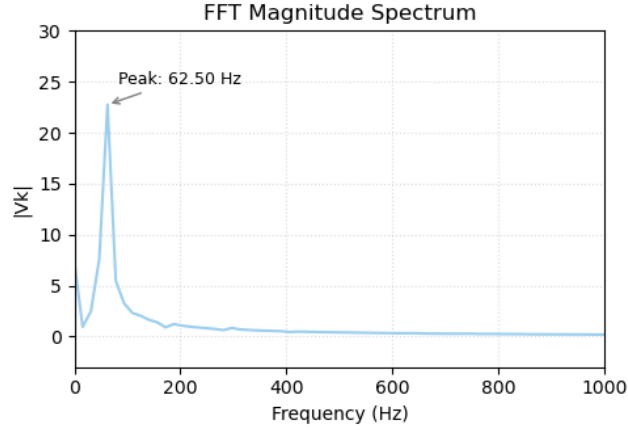


Figure 5: FFT magnitude spectrum showing the fundamental at 62.5 Hz and harmonic distortion.

We used a nonlinear least-squares fit to the sine output to get the following parameters: $A = 25.02$ V, $f = 58.16$ Hz, $\phi = -0.014$ rad, and $V_0 = 4.62$ V. The computed RMS voltage of 17.69 V was close to the measured 17.25 V, confirms the waveform's sinusoidal nature. Figure 6 shows the original waveform and fit.

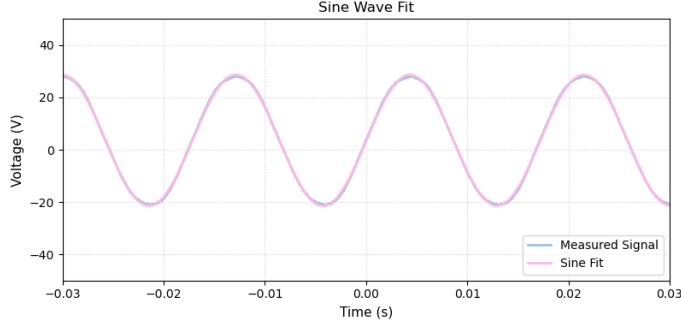


Figure 6: Nonlinear least-squares fit to the sine wave output.

The recorded frequency from the oscilloscope was approximately 58.08 Hz, is close to the fitted value of 58.16 Hz and is within reasonable error of FFT-detected 62.5 Hz fundamental. These small differences are likely due to resolution in the FFT bins and oscilloscope sampling rate. We also observed minor deviations during longer runtimes, which could affect waveform symmetry and stability. Despite these factors, the generator produced consistent outputs across all stages, and our results aligned closely with theoretical expectations.

3 Conclusion

In this project, we successfully designed and built a function generator circuit capable of producing square, triangle, and sine waveforms using op-amp stages and passive shaping components. The comparator-inverter loop reliably generated a square wave, which was integrated into a triangle waveform and subsequently converted into a sine wave using a diode resistor network and summing amplifier. Our measurements showed consistent waveform frequency across all stages. The output sine wave had a measured frequency of approximately 58.08 Hz, with the nonlinear curve yielding 58.16 Hz and FFT-based analysis detecting a 62.5 Hz peak. The total harmonic distortion was 11.44%, which indicates moderate deviation from a pure sine likely due to diode nonidealities, waveform asymmetries, and component tolerances.

Small discrepancies between computed and measured frequencies may have resulted from oscilloscope sampling limitations, FFT bin resolution, and phase noise in the signal. Additionally, the slight mismatch in our power supply rails (e.g., ± 17.98 V vs. ± 18.00 V) resulted in asymmetry that likely affected integration slopes and sine shaping. Additionally, component tolerances (especially resistors in the feedback paths) and thermal drift over time also contributed to deviation from ideal behavior [7].

Future improvements could include using tighter-tolerance components, adding trimmer potentiometers for threshold calibration, and incorporating active fil-

ter stages to better suppress harmonics [4]. Refinement of waveform quality can also be tested by adjusting component values such as R_2 to modify the integrator slope and output frequency, or by fine-tuning the supply voltage $V_{\pm}$ to control the amplitude. Finally, we suggest improving power supply regulation and thermal compensation to improve stability and reduce long-term drift.

Acknowledgements

We would like to thank our laboratory instructor, Dr. Heinzen, and our teaching assistant, Kyle Gable, for their guidance and support through these labs. Their assistance and feedback were invaluable in helping us understand the experimental procedures and circuit concepts. We also appreciate the department's laboratory facilities and equipment, which made these experiments possible.

References

- [1] Anna Bouchaleun. An elementary introduction to fast fourier transform algorithms. 2019.
- [2] EEE Guide. Triangular wave generator using op amp, 2016. Accessed: 2025-12-07.
- [3] Rod Elliott. Sinewave oscillators - characteristics, topologies, and examples, 2010. Accessed: 2025-12-07.
- [4] Daniel Fallows, Stefano Nuzzo, Alessandro Costabeber, and Michael Galea. Harmonic reduction methods for electrical generation: a review. *IET Generation, Transmission & Distribution*, 12(13):3107–3113, 2018.
- [5] Daniel Heinzen. Digital oscilloscopes, signal processing, and spectrum analysis, v.i., 2025. Physics 338K, Fall 2025.
- [6] Paul Scherz and Simon Monk. *Practical Electronics for Inventors, Fourth Edition*. McGraw-Hill, 2016.
- [7] Mohamed S. Seddik, Magdy B. Eteiba, and Jehan Shazly. Evaluating the harmonic effects on the thermal performance of a power transformer. *Energies*, 17(19), 2024.
- [8] David Williams. Understanding, calculating, and measuring total harmonic distortion (thd). *All About Circuits*, February 2017. Accessed: 2025-12-07.